WEEK 2 – LIBRARY AUTOMATION

INTRODUCTION TO LIBRARY AUTOMATION

Meaning of Library Automation

Library automation refers to the **application of computers, software, networking technologies, and other ICT tools** to perform traditional library functions **faster, more accurately, and more efficiently**.

In simple terms, library automation means **doing library work with machines instead of manual methods**.
This includes cataloguing, classification, circulation, acquisitions, serial control, reference service, and other operations.

Automation does *not* change the objectives of the library; instead, it **improves how those objectives are achieved**.

Automation also reduces redundancy, human error, delays, and the physical stress involved in paper-based operations.

HISTORICAL DEVELOPMENT OF LIBRARY AUTOMATION

Early Efforts (Before Computers)

Before computers were introduced, libraries used mechanical tools to speed up operations:

* Typewriters

* Kardex systems (for serials control)

* Microfilm and microfiche readers

* Manual card catalogues

These were not fully automated but served as stepping stones.

First Phase: 1950s – 1960s

This marks the **beginning of computer use** in libraries.
Large academic and research libraries in the US began experimenting with:

* Punch-card systems

* Mainframe computers

* Machine-readable cataloguing (MARC)

LIS211 NOT FOR SALE

MARC, introduced by the Library of Congress, became a **global standard** for representing bibliographic information in electronic form.

Second Phase: 1970s – 1980s

This phase brought:

* Growth of library networks

* Shared cataloguing

* Emergence of Online Public Access Catalogues (OPAC)

* Introduction of integrated library systems (ILS)

Libraries began moving away from card catalogues.

Third Phase: 1990s – 2000s

Major developments:

* Internet adoption

* Web-based catalogues

* Digital libraries

* Electronic resources

* Institutional repositories

Current Phase (2010s – Present)

Modern library automation focuses on:

* Cloud-based library systems

* Mobile library applications

* Self-service technologies

* Artificial intelligence

* RFID for circulation

* Big data and analytics

LIS 211 NOT FOR SALE

* E-resource management systems

Automation is now a **core expectation**, not an upgrade.

REASONS FOR LIBRARY AUTOMATION

1. To Improve Efficiency

Automation speeds up routine tasks such as cataloguing, issuing books, and returning items.

2. To Reduce Human Error

Computers are more consistent and reliable.

3 To Enhance User Satisfaction

Automated libraries provide:

* Quick search

* Accurate information

* Faster access to resources

4 To Handle Increasing Information Resources

Modern libraries manage huge volumes of information that manual systems cannot handle.

5 To Enable Remote and 24/7 Access

Students and researchers can access the catalogue anytime.

6 To Support Digital Transformation

Automation is necessary for:

* E-books

* Databases

* Digital archives

* Online journals

7 To Enable Resource Sharing

Libraries can share catalogues and resources through networks and consortia.

LIS 211 NOT FOR SALE

8 To Improve Management and Decision-Making

Automated systems generate statistics and reports for planning.

AREAS OF LIBRARY OPERATIONS THAT CAN BE AUTOMATED

This section is very important. Expand fully:

1 Cataloguing and Classification

Automation simplifies:

* Creation of bibliographic records

* Importing MARC records

* Editing records

* Copy cataloguing

* Searching subject headings

* Assigning classification numbers

* It also reduces duplication of work.

2 Circulation Control

Automation manages:

* Charging and discharging of books

* Renewals

* Reservations

* Overdue notices

* User registration

* Fine calculation

* Barcodes and RFID make circulation faster and more accurate.

LIS 211 NOT FOR SALE

3 Acquisition

Automation helps to:

* Create purchase orders

* Track orders

* Maintain vendor records

* Receive materials

* Manage budgets

* It improves accountability and financial reporting.

4 Serials/Periodicals Control

Automating serials helps libraries to:

* Track journal arrivals

* Manage subscriptions

* Claim missing issues

* Maintain holdings

* Generate statistics

* This is important because serials are time-sensitive.

5 OPAC (Online Public Access Catalogue)

OPAC replaces card catalogues and allows users to:

* Search the library catalogue online

* Check availability of items

* View new arrivals

* Place reservations

* Modern OPACs may support:

    * Keyword search

LIS 211 NOT FOR SALE

* Boolean search

* Author/title/subject search

6 Reference Services

Automation supports:

* Online reference queries

* Virtual reference chats

* Databases

* Subject guides

* E-mail reference

* FAQ systems

7 Digital Libraries and Institutional Repositories

Automation supports:

* Publication storage

* Thesis and dissertation management

* Metadata creation

*Long-term digital preservation

8. Management and Statistics

Automated systems generate reports on:

* Most used books

* Peak borrowing times

* Departmental usage

* Annual statistics

* This supports evidence-based decision-making.

LIS 211 NOT FOR SALE

5. COMPONENTS OF LIBRARY AUTOMATION

Automation consists of multiple components working together:

Hardware

These are physical devices used in library automation:

* Computers

* Servers

* Printers

*Bar-code scanners

* RFID readers and gates

* Photocopiers

* Scanners

* UPS and stabilisers

* Networking devices (routers, switches)

Software

Library software is divided into:

a. System Software

Operating systems such as Windows, Linux, macOS.

b. Application Software

Integrated Library Management Systems (ILMS), for example:

1. Koha
2. Evergreen
3. Virtua
4. Alice for Windows
5. Destiny
6. SLAM
7. LibPlus

LIS 211 NOT FOR SALE

Databases

Every automated library depends on databases for storing:

1. Bibliographic records
2. User records
3. Transaction logs
4. Holdings

Common database systems include:

* MySQL

* PostgreSQL

* Oracle

* SQL Server

Networks

Networked systems enable:

1. Resource sharing
2. OPAC access
3. Remote services
4. Intranet communication

Networks can be:

* LAN

*WAN

* Wi-Fi

* Campus networks

People

Humans remain critical:

1. Librarians
2. System administrators
3. ICT technicians
4. Database managers

LIS 211 NOT FOR SALE

5. Vendors and trainers
6. Users

Automation only works when people understand how to use the system.

STEPS IN IMPLEMENTING LIBRARY AUTOMATION

A detailed explanation for teaching:

Needs Assessment

The library must determine:

1. Why automation is needed
2. What problems exist
3. What goals need to be achieved

Feasibility Study

Examines:

* Technical feasibility

* Financial feasibility

* Human resource capacity

* Infrastructure availability

Selecting the Modules to Automate

No library automates everything at once. They prioritise:

1. Cataloguing
2. Circulation
3. OPAC
4. Acquisition

Choosing the Right Software

Consider:

1. Cost
2. Customisability
3. User-friendliness
4. Vendor support
5. Compatibility with standards (MARC, Z39.50)
6. Integration with existing systems

LIS 211 NOT FOR SALE

Staff Training

Training ensures librarians know:

1. How to operate the system
2. How to troubleshoot common problems
3. How to assist users
4. Without training, automation fails.

Data Conversion

This is often the most time-consuming part:

1. Retrospective conversion of card catalogues
2. Scanning and digitisation
3. Data cleaning

System Installation and Testing

Test for:

1. Functionality
2. Speed
3. Reliability
4. Errors
5. Network stability

Full Deployment

System goes live. Library now operates with the new software.

Maintenance and Continuous Updates

Automation requires:

1. Regular backups
2. Software updates
3. Hardware replacement
4. Technical support

CHALLENGES OF LIBRARY AUTOMATION

1. Inadequate Funding

Many libraries in developing countries lack sufficient budgets.

2. Poor ICT Infrastructure

Unstable power supply, bad networks, and insufficient hardware.

LIS 211 NOT FOR SALE

3. Lack of Skilled Manpower

Shortage of trained ICT librarians and technicians.

4. Resistance to Change

Some staff prefer manual systems and fear technology.

5. High Cost of Software

Proprietary systems are expensive.

6. Sustainability Issues

After initial implementation, maintenance is often neglected.

BENEFITS OF LIBRARY AUTOMATION

Make this section strong:

1. Faster retrieval of information
2. Improved accuracy and consistency
3. Increased productivity
4. Enhanced user experience
5. Reduced workload
6. Efficient management of resources
7. Easy sharing of bibliographic records
8. Availability of online services
9. Better preservation of materials
10. Greater transparency and accountability

LIS 211 NOT FOR SALE